

CA-PQ Yellow Paper

Formal Specification

CA-PQ Project

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Normative companion to code. Where this paper and code disagree, code governs and the paper is a bug—file it.

Abstract

Notation: $\parallel$ is concatenation, bytes are little-endian unless stated, $H(c, ctx)$ is keyed BLAKE3, $XOF(data, ctx, n)$ is keyed BLAKE3-XOF. All integers in key paths are exact (no floats anywhere). Test vectors live in-crate (`kat_*` tests) and in the independent Python transcriptions kept outside the tree.

Contents

1	Conventions and sizes	2
2	Chaotic engine	2
3	Hashing	2
4	X25519	3
5	X-KEM	3
6	ML-KEM	3
7	Hybrid combiners	4
8	AEAD	4
9	Domain registry	4
10	KAT inventory	4
11	Conformance notes	5

1 Conventions and sizes

Security parameter: 256-bit seeds throughout; shared secrets 32 bytes. Wire integers have fixed sizes by type (no length prefixes on wire objects). Sysids (4 bytes, bound into commits): X-KEM PQ04, hybrid-768 PQH2, hybrid-512 PQH1.

Object	X-KEM	ML-KEM-768	ML-KEM-512	Hybrid-768	Hybrid-512
Public key	68	1 184	800	1 284	900
Secret key	64 (32 seed)	2 400	1 632	2 464 (32 seed)	1 696 (32 seed)
Ciphertext	60 = 12+32+16	1 088	768	1 148	828
Shared secret	32	32	32	32	32

AEAD tag: 16 bytes. KEM/hybrid nonce: 12 bytes.

2 Chaotic engine

Fixed-point Q32.32 (FRAC = 32) in i64, i128 intermediates. $DT_Q = 4294967$ (round($0.001 \cdot 2^{32}$)), $N_{\text{TRANSIENT}} = 1000$, clamp bound $\text{BOUND}_Q = 256 \cdot 2^{32}$, sampling STRIDE = 10, extractor scale 2^{20} .

State $s = [x, y, z, w]$, params $p = [a, b, c, d]$, defaults $a = 10$, $b = 8/3$, $c = -1$, $d = 28$ (Lorenz-28 regime).

Derivative with $\text{qmul}(u, v) = (u \cdot v) \gg 32$ (exact truncating):

$$\begin{aligned} dx &= a \cdot (y - x) + w/4 \\ dy &= d \cdot x + c \cdot y - x \cdot z \\ dz &= x \cdot y - b \cdot z \\ dw &= y/8 - w/4 - x/16 \end{aligned}$$

RK4 step (fixed dt): $k_1 = f(s)$, $k_2 = f(s + k_1 \cdot dt/2)$, $k_3 = f(s + k_2 \cdot dt/2)$, $k_4 = f(s + k_3 \cdot dt)$, $s += \text{qmul}(\text{div6}(k_1 + 2k_2 + 2k_3 + k_4), dt)$ where div6 is bit-exact truncating $/6$ (Granlund–Montgomery, range-checked $|v| < 2^{50}$). All advances clamped to $\pm \text{BOUND}_Q$. $\text{iterate}(s, p, n)$ applies n steps with $dt = DT_Q$.

Expansion (`expand_from_bytes`, 64 B $\rightarrow$ state + params): bytes $[0..32] \rightarrow x_0 \in [-10, 10]$ per coordinate via centered u32, Q-shift, $\times 10$, fractional mixing from bytes $[32..36]/[48..52]$; params = defaults jittered $\pm 5\%$ from bytes $[36..40]$ ($\text{delta} = p \cdot j/2560$, $j \in [-128, 127]$). Bytes 16–31, 40–47, 52–63 are unused by the map (lossy, harmless).

Extraction: with $u = x$ as u64, $\text{folded} = (u \oplus (u \gg 17) \oplus (u \gg 29) \oplus (u \gg 7)) \& 0\text{xFF}$, $\text{out} = \text{folded} \oplus \text{rotr}_3((u \gg 11) \& 0\text{xFF}) \cdot 0\text{x9E} \text{ rotr}_2$. **keystream** emits $\text{extract}(x_0) \oplus \text{extract}(x_1) \oplus \text{extract}(x_3)$ every STRIDE steps (transient assumed discarded by caller).

Health (`health_check`, validation-only): 1000-step boundedness + movement (drift $\geq 2^{-8}$) + 1024-step 1-LSB divergence ($\geq 8/128$ low-word diffs).

3 Hashing

BLAKE3: standard IV, message permutation $[2, 6, 3, 10, 7, 0, 4, 13, 1, 11, 12, 5, 9, 14, 15, 8]$, 7-round compress, tree mode (`BLOCK_LEN` = 64, `CHUNK_LEN` = 1024, `MAX_DEPTH` =

54, chunk counter, `CHUNK_START/CHUNK_END/ROOT` flags). API: `Hasher::new, new_keyed, update, finalize, finalize_xof`, `blake3, blake3_keyed, derive_key, Hash256::hash, xof, mac`.

Keccak sponge (FIPS 202): compile-time LFSR round constants, $\rho/\pi/\chi/\iota$ reference layout, `pad10*1`, suffixes `0x06` (SHA3) / `0x1F` (SHAKE). Rates: SHA3-256/SHAKE256 136, SHA3-512 72, SHAKE128 168. API: one-shot `sha3_256/512, shake128/256`; incremental and streaming readers; 4-way batch squeezes.

SIMD (`chaos-simd`, compile-time dispatch, no runtime detection): BLAKE3 compress 8-way AVX2 / 4-way SSE2 / 4-way NEON; Keccak-f 4-way AVX2 / 2-way SSE2 / 2-way NEON; NTT 8-way AVX2 / 4-wide NEON (+ portable everywhere). Hasher groups narrow $8 \rightarrow 4$ chunks on SSE2/NEON tiers with identical tree semantics.

4 X25519

Field $GF(2^{255} - 19)$ in 5×51 -bit signed i64 limbs, i128 products; full + light carry with 19-fold; schoolbook multiply. Montgomery ladder (255 steps, branchless cswap), $A24 = 121665$, inversion $z^{2^{255}-21}$ by fixed chain, full encode/decode with commit reduction. Scalar clamping (&248, &127, |64), point masking (&0x7f), all-zero contributory check.

5 X-KEM

Sysid PQ04.

keygen(seed32)

$x_{\text{sec}} = \text{XOF}(\text{seed}, \text{ca-pq-sk-x-v1})$, $\text{chaos_seed} = \text{XOF}(\text{seed}, \text{ca-pq-sk-chaos-v1})$, $x_{pk} = \text{scalarmult}(x_{\text{sec}}, G)$, $\text{mat} = \text{XOF}(\text{chaos_seed}, \text{ca-pq-kdf-v1})$, $\text{commit} = H(\text{PQ04} \parallel \text{mat} \parallel x_{pk}, \text{ca-pq-pk-v1})$.

encaps(pk, eph_seed32, nonce12)

$\text{eph}_x = \text{XOF}(\text{eph_seed}, \text{ca-pq-eph-x-v1})$, eph_{pk} , $\text{shared} = \text{DH}(\text{eph}_x, \text{pk} \cdot x_{pk})$ (None if zero), $\text{tr} = \text{commit} \parallel x_{pk} \parallel \text{eph}_{pk} \parallel \text{nonce}$, $\text{tr}_{\text{hash}} = H(\text{tr}, \text{ca-pq-tr-v1})$, $\text{mac}_{\text{key}} = H(\text{shared}, \text{ca-pq-mac-v1})$, $\text{tag} = \text{MAC}(\text{mac}_{\text{key}}, \text{tr}, \text{ca-pq-ct-v1})[..16]$, $\text{tail} = \text{keystream}(\text{expand}(\text{XOF}(\text{shared} \parallel \text{tr}_{\text{hash}}, \text{ca-pq-chaos-v1})))$, $\text{ss} = H(\text{tail} \parallel \text{shared} \parallel \text{tr}_{\text{hash}}, \text{ca-pq-sync-v1})$. Ciphertext = nonce || eph_pk || tag.

decaps(sk, pk, ct)

Recompute shared (None if zero), verify tag (abort $\rightarrow$ None), recompute tail + ss identically.

6 ML-KEM

Parameters: $n = 256$, $q = 3329$, $\text{INV128} = 3303$; 768: $k = 3$, $\eta_1 = \eta_2 = 2$; 512: $k = 2$, $\eta_1 = 3$, $\eta_2 = 2$; both $d_u = 10$, $d_v = 4$. Generic core over Profile $\{K, \text{ETA1}, \text{ETA2}, \text{EK_LEN}, \text{DK_LEN}, \text{CT_LEN}\}$.

NTT: 7 layers (len $128 \rightarrow 2$ fwd / $2 \rightarrow 128$ inv), twiddles $17^{\text{brv7}(i)}$ (brv7 = 7-bit reversal), $\text{fwd_idx}(\text{len}, g) = 256 / (2 \cdot \text{len}) + g$, pair moduli $\gamma[p] = 17^{\text{brv7}(64 + \lfloor p/2 \rfloor) + 128 \cdot (p \bmod 2)}$, single final $\times 3303$ scale. Modular ops: branchless conditional add/sub on $[0, q)$, single-% mulmod; entry accepts $[0, 4096)$ (decode range) via one conditional subtract—scalar and vector paths agree bit-for-bit there.

Codec: LSB-first `byte_encode/decode`; $\text{compress}_d(x) = \text{round}(2^d / q \cdot x) \bmod 2^d$ / $\text{decompress}_d(y) = \text{round}(q / 2^d \cdot y)$ (exact integer half-up); `cbd $_{\eta}$` ($\eta = 2$: 128 B, $\eta = 3$: 192 B); `sample_ntt(seed, i, j)`

= SHAKE128(seed|| i || j) rejection sampling (3-byte parse, $d_1 = b_0 + 256 \cdot (b_1 \bmod 16)$, $d_2 = b_1/16 + 16 \cdot b_2$); $\text{prf}_\eta = \text{SHAKE256}(s||b, 64 \cdot \eta \text{ B})$.

Flows (FIPS-203 shape): $\rho||\sigma = \text{SHA3-512}(d)$; matrix from ρ , noise from σ (counters $0..2k-1$, η_1); **encaps**: coins $0..2k$ (η_1) + e_2 (η_2), FO $K = H(m||H(ek))$, $ss = \text{SHAKE256}(K||H(ct))$; **decaps**: decrypt, re-encrypt, constant-time select vs. $\text{SHAKE256}(z||H(ct))$ implicit rejection. **encaps_batch**: shared matrix, element i bit-identical to scalar encaps. Encode widths $12/10/4/1$; **message_poly** maps bits to $(q+1)/2$.

7 Hybrid combiners

Both profiles: subseeds $H(\text{seed}, \text{h-x/h-mlkem-d/h-mlkem-z})$ (768: **ca-pq-h-***, 512: **ca-pq-h512-***); $\text{commit} = H(xpk.\text{commit}||xpk.\text{sysid}||xpk||pqek, \text{h-pk})$; encaps derives eph_x , m_{pq} from seed, runs both legs, $tr = H(\text{commit}||ct_x||ct_{pq}||\text{nonce}, \text{h-tr})$, $ss = H(ss_x||ss_{pq}||tr, \text{h-ss})$. Decaps: X failure $\rightarrow$ None; PQ tamper $\rightarrow$ unrelated ss . Fixed sizes per §1; sysids PQH2/PQH1.

8 AEAD

Stream: keyed XOF over length-framed nonce (**ca-pq-dem-stream-v2**: domain || 0x00 || le64(len) || nonce—full nonce absorbed, no truncation). Tag: $\text{MAC}(ss, \text{ca-pq-dem-v1}||0x00||\text{nonce}||ct)[..16]$, constant-time verify, verify-before-decrypt. v1 streams ($ss||\text{nonce}[..24]$ zero-padded) are incompatible by design (collided distinct nonces).

9 Domain registry

Each string is used for exactly one purpose (duplicates below are same-value-both-sides by construction):

- Key derivation/splitting: **ca-pq-sk-chaos-v1**, **ca-pq-sk-x-v1**, **ca-pq-kdf-v1**, **ca-pq-h-x**, **ca-pq-h-mlkem-d/z**, **ca-pq-h512-x/mlkem-d/mlkem-z**, **ca-pq-h-eph-x/m**, **ca-pq-h512-eph-x/m**.
- Commits/transcripts/MACs: **ca-pq-pk-v1**, **ca-pq-tr-v1**, **ca-pq-mac-v1**, **ca-pq-ct-v1**, **ca-pq-sync-v1**, **ca-pq-chaos-v1**, **ca-pq-eph-x-v1**, **ca-pq-h-pk/tr/ss**, **ca-pq-h512-pk/tr/ss**.
- DEM: **ca-pq-dem-stream-v2**, **ca-pq-dem-v1**.
- ML-KEM internals use raw SHA3/SHAKE (FIPS-assigned roles, no custom domains).

10 KAT inventory

BLAKE3 official vectors + split/multi-block XOF tests (**chaos-hash**); SHA3/SHAKE oracle incl. rate boundaries; X25519 big-int goldens + field-layer goldens (**x25519::tests**); ML-KEM NTT/CBD/compress goldens + full **kat_a/kat_b** vs. independent transcriptions (both profiles) + roundtrip/tamper/batch suites; hybrid **kat_hybrid_a/b** + **kat_h512_a/b** vs. full-stack references; chaos **kat_state** + health/avalanche gates; AEAD roundtrip/tamper/framing tests; attack-suite gates (**chaos-attack**).

11 Conformance notes

Implementations must reproduce: exact fixed-point/RK4/div6 semantics (§2), NTT twiddle order and pair moduli (§6), FO domain flow, wire sizes (§1), and domain strings (§9). Deviating in constant-time hardening is allowed only where outputs are unaffected. No interoperation with other libraries is claimed; agreement is defined against the in-tree KATs and the independent Python transcriptions.